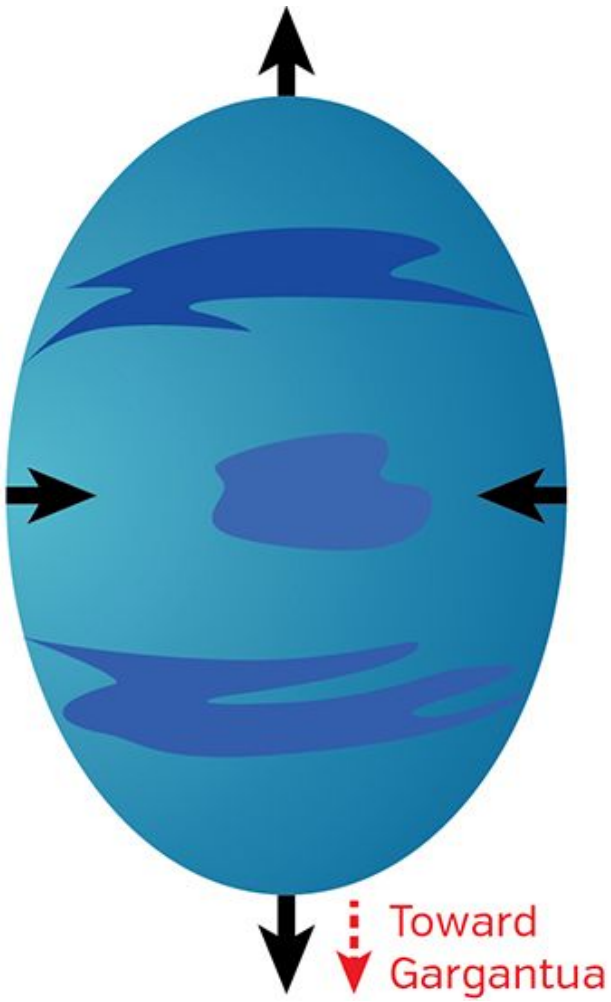



*Fig. 17.3.* Tidal deformation of Miller's planet.

If Miller's planet were to rotate relative to Gargantua (if it didn't keep the same face toward Gargantua at all times), then as seen by the planet, the tidal forces would rotate. First the planet would be crushed east-west and stretched north-south. Then, after a quarter rotation, the crush would be north-south and the stretch east-west. These crushes and stretches would be enormous compared to the strength of the planet's mantle (its solid outer layers). The mantle would be pulverized, and then friction would heat it and melt it, making the whole planet red hot.

That's not at all what Miller's planet looks like! So the conclusion is clear: In my science interpretation, the planet must always keep the